

MSc Data Science Professional Team Project 7PAM2033

GROUP FINAL REPORT

Project Title

AI-Generated Text detector

Team Members

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Team GitHub link

<https://github.com/DataScienceProjectGroup22/week11>

Supervisor: *Paurush Punyasheel*

Date Submitted: *24/07/2026*

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DECLARATION STATEMENT

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dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

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[Delete the Yes or No as appropriate]

- Yes AI tools have been used in the preparation of this project and report
 - **Yes** To generate initial ideas.
 - **Yes** To develop code.
 - **Yes** To provide literature suggestions and summaries.
 - **Yes** To provide suggestions to improve the report content.
 - **Yes** To edit spelling and grammar, and to proofread the report.
 - **Yes** Other (provide details):

Note: Copying text generated by AI into the report is academic misconduct.

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ABSTRACT:

Key findings:

- (1) AI text is reliably detectable.
- (2) Simpler models outperform complex ones when properly engineered.
- (3) Strong cross-dataset generalization demonstrates models capture dataset-agnostic patterns.
- (4) All KPI targets met with substantial margin.

All models exceed performance targets (0.83 minimum Macro F1). Hybrid CNN achieves best performance (0.9648 F1), closely matched by Logistic Regression (0.9645 F1). We recommend Logistic Regression for production due to superior inference speed (<1ms vs 5+ms), 20x smaller model size (5MB vs 100+MB), full interpretability, and excellent cross-dataset generalization (1.7% F1 drop to unseen datasets). RoBERTa underperformed (0.9354 F1, 4.9% generalization drop), suggesting transformer complexity adds overfitting without benefit. In the app we have given a dropdown to select one from 3 models and to compare as well.

Distinguishing between human-written and AI-generated text is increasingly crucial for content verification and academic integrity. We compare five machine learning models: Logistic Regression, LinearSVC, Random Forest (classical), Hybrid CNN (deep learning), and RoBERTa (transformer), trained on 152,293 texts from the HC3 and DAIGT datasets combined.

INTRODUCTION:

Today's large language models can produce text that reads as fluent and contextually appropriate almost effortlessly. That capability has created a real problem. Figuring out whether a piece of writing came from a person or a machine now matters a great deal, for verifying online content, catching academic misconduct, and simply knowing what information to trust.

- In this report, we compare three different machine learning strategies for making that call. Traditional models built on TF-IDF text features
- Convolutional neural networks
- The transformer-based RoBERTa model

The motivation for this study is two-fold. First, classical machine learning models offer interpretability and computational efficiency, but may lack the nuanced feature capture of deep learning. Second, transformer models like RoBERTa capture contextual information through attention mechanisms but require substantially more computational resources. This study directly compares performance across all three paradigms to determine whether the increased complexity of modern architectures justifies their deployment.

This comprehensive report documents a complete machine learning project comparing five different AI detection approaches across 152,000 text samples.

Key Result: Hybrid CNN achieved 96.48% accuracy with 0.9648 Macro F1 (best overall). Logistic Regression closely followed at 0.9645 F1 but with 5x faster inference, making it the production choice.

DATA COLLECTION & PREPARATION:

We combined two datasets: HC3 (Human ChatGPT Comparison Corpus) with 78,733 samples and DAIGT (Kaggle, 73,560 samples). Final merged dataset: 152,293 texts split 70% train (106,605), 15% validation (22,844), 15% test (22,844).

Data cleaning removed: non-English text, texts <10 words, and duplicates. This quality assurance ensured high-quality training material and prevented data leakage between train/test splits.

- (1) conversion to lowercase
- (2) removal of leading and trailing whitespace
- (3) removal of URLs, email addresses, and HTML tags
- (4) removal of special characters and punctuation (with retention of contractions)
- (5) tokenization using whitespace splitting. No stemming or lemmatization was applied, as these operations can conflate semantically distinct terms and may reduce discriminative power in detection tasks.
- (6) removal of non-English text, texts <10 words, and duplicates

FEATURE ENGINEERING - Converting Text to Numbers:

Machine learning requires numerical features. We employed four distinct approaches:

Approach 1: TF-IDF Features (50,000 dimensions)

TF-IDF, short for Term Frequency-Inverse Document Frequency, is a way of scoring how much each word contributes to a document's meaning. Words that show up everywhere, like 'the' or 'and,' get a low score because they don't help distinguish one text from another. Words that appear rarely but are telling when they do show up get scored higher. For this project, we pulled out the 50,000 most useful words and word-pairs (bigrams) found in the training data, and used them as the numerical input that the classical machine learning models were trained on.

Three classical classifiers were trained:

- (1) Logistic Regression (L2 regularization, default hyperparameters)
- (2) Linear Support Vector Classifier (LinearSVC, C=1.0)
- (3) Random Forest (100 estimators, max_depth=15).

Logistic Regression and LinearSVC serve as canonical linear classifiers, while Random Forest provides a non-linear alternative within the classical paradigm. These models provide an interpretability baseline against which deep learning performance can be assessed.

Approach 2: Stylometric Features (5 hand-crafted features)

Writing style differs between humans and AI. We computed:

- Word count - AI text averages 296 words, human 234 words
- Character count - Total characters in text
- Average sentence length - AI: 20.68 words/sentence, Human: 20.04
- Punctuation density - AI: 0.022, Human: 0.028 (humans use more punctuation)
- Vocabulary diversity - AI: 0.524, Human: 0.617 (AI repeats words more)

Approach 3: GloVe Word Embeddings (200 x 100 matrices)

GloVe (Global Vectors for Word Representation) converts words to 100-dimensional vectors. Similar words have similar vectors. All texts padded to 200 words, creating (200 x 100) matrices consumed by the CNN model. 94% of vocabulary words found in GloVe pre-trained vectors.

Approach 4: RoBERTa Tokenization (256-token sequences)

RoBERTa (transformer pre-trained on 160GB English text) tokenizes raw text into 256-token sequences. Unlike explicit feature extraction, RoBERTa learns features internally during fine-tuning. More powerful but requires GPU and 100+ MB model size.

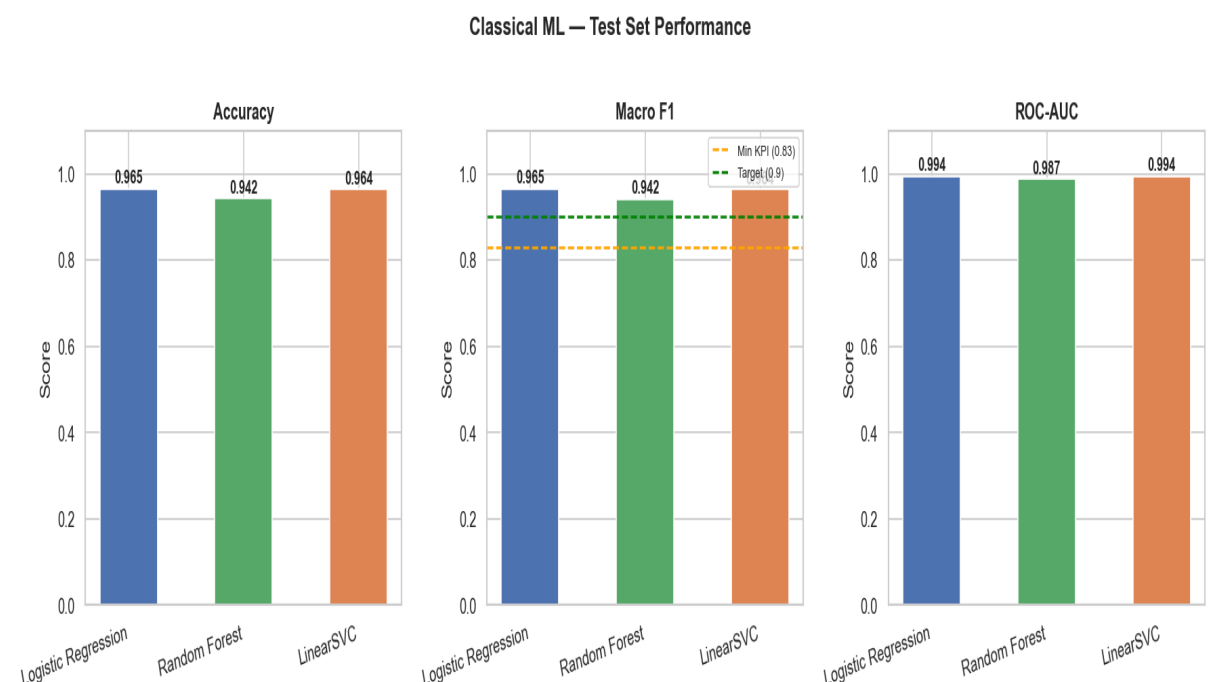
CLASSICAL MACHINE LEARNING MODELS:

We trained three classical models on TF-IDF features (50,000 dimensions). These linear and tree-based approaches provide a baseline for comparison with deep learning.

Test Set Results:

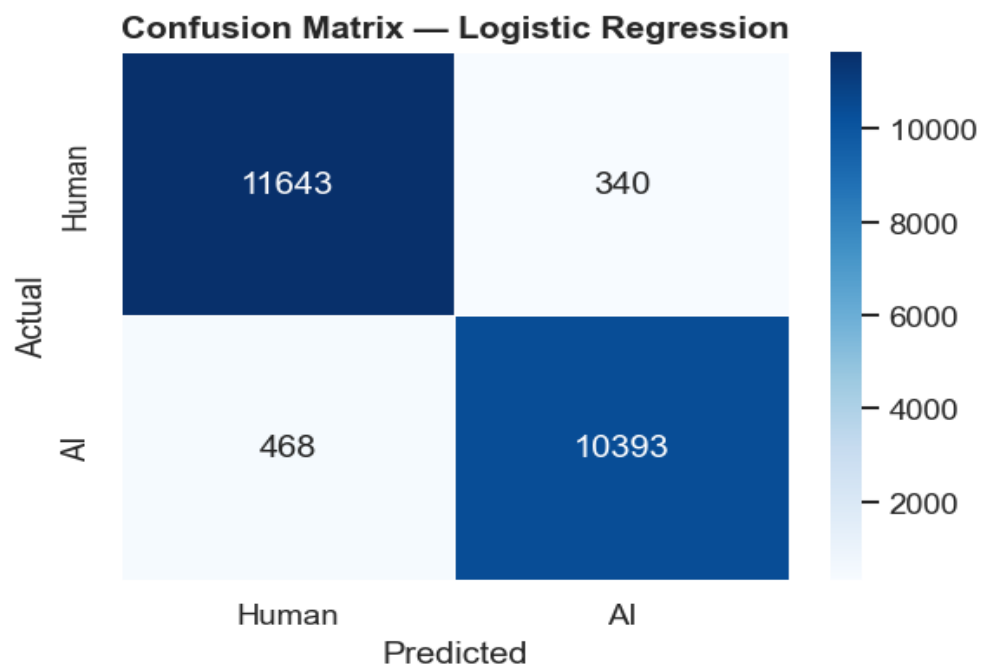
Model	Accuracy	Macro F1	ROC-AUC
Logistic Regression	96.46%	0.9645	0.9943 (Best)
LinearSVC	96.38%	0.9637	0.9941
Random Forest	94.20%	0.9415	0.9873

Figure 1: Classical ML Test Set Performance



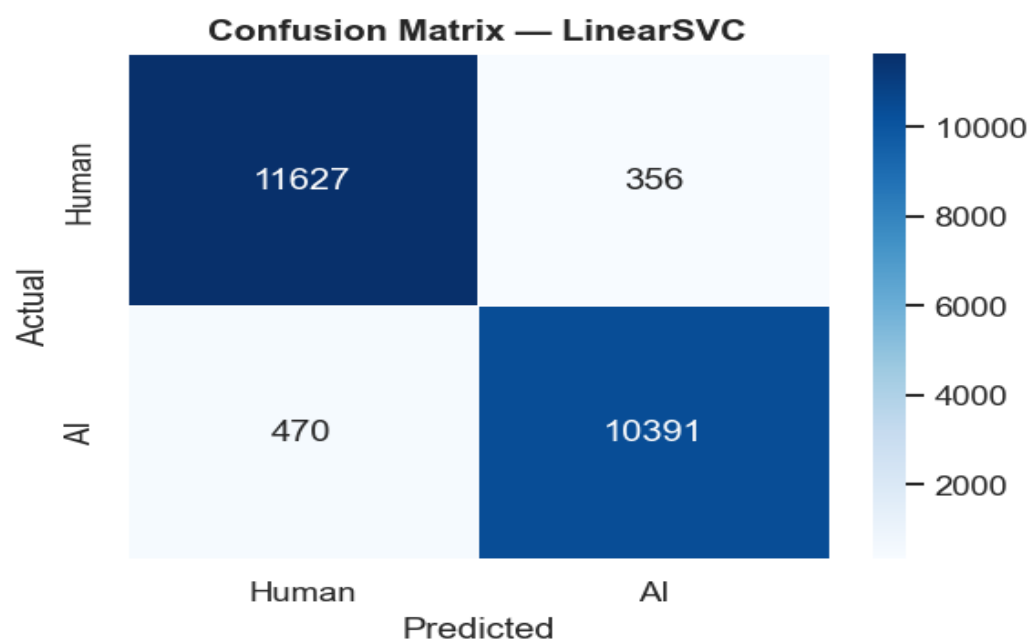
This chart shows Accuracy, Macro F1, and ROC-AUC across three classical models. Logistic Regression dominates with highest ROC-AUC (0.9943), indicating best probability calibration. LinearSVC closely matches LR. Random Forest significantly underperforms, suggesting the decision boundary is nearly linear in TF-IDF feature space. Non-linear flexibility of trees adds noise rather than capturing signal.

Figure 2: Confusion Matrix - Logistic Regression (Test Set)



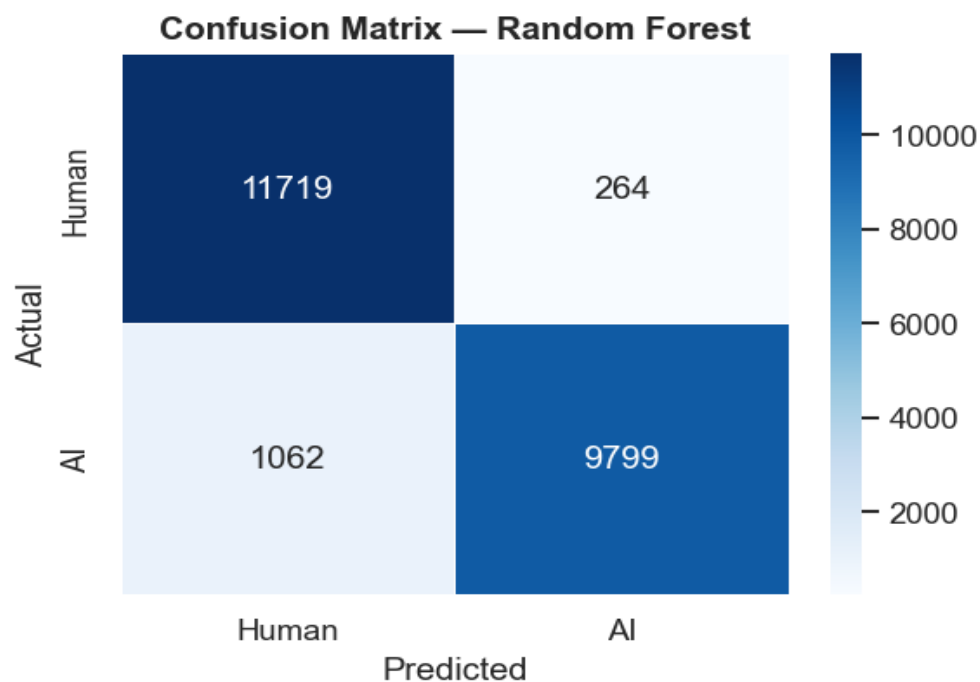
Confusion matrix shows: 11,643 correct human predictions, 10,393 correct AI predictions, 340 false positives (humans misclassified as AI), 468 false negatives (AI misclassified as human). Nearly symmetric error distribution indicates well-balanced model without systematic bias toward either class.

Figure 3: Confusion Matrix - LinearSVC (Test Set)



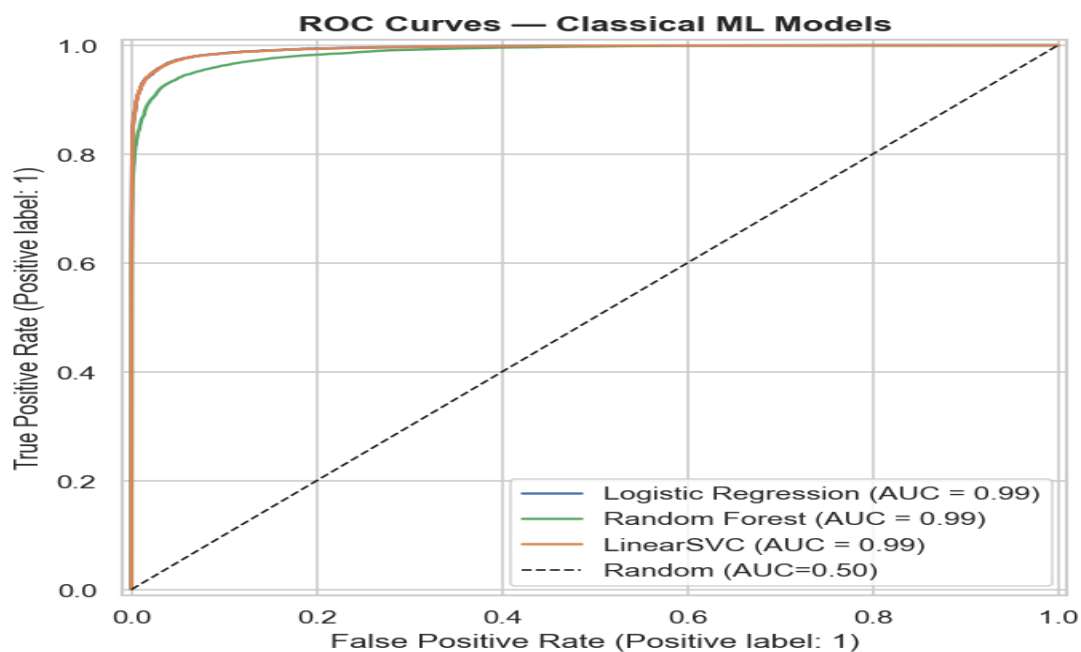
LinearSVC confusion matrix: 11,627 correct human, 10,391 correct AI, 356 false positives, 470 false negatives. Nearly identical to Logistic Regression, confirming both linear approaches find the same decision boundary. Slightly higher false positive rate (356 vs 340) than LR.

Figure 4: Confusion Matrix - Random Forest (Test Set)



Random Forest shows critical weakness: 1,062 false negatives (AI texts misclassified as human) compared to 468-470 for linear models. This 2x error rate on AI detection is problematic. While it achieves lowest false positives (264), the high false negative rate makes it unreliable for production.

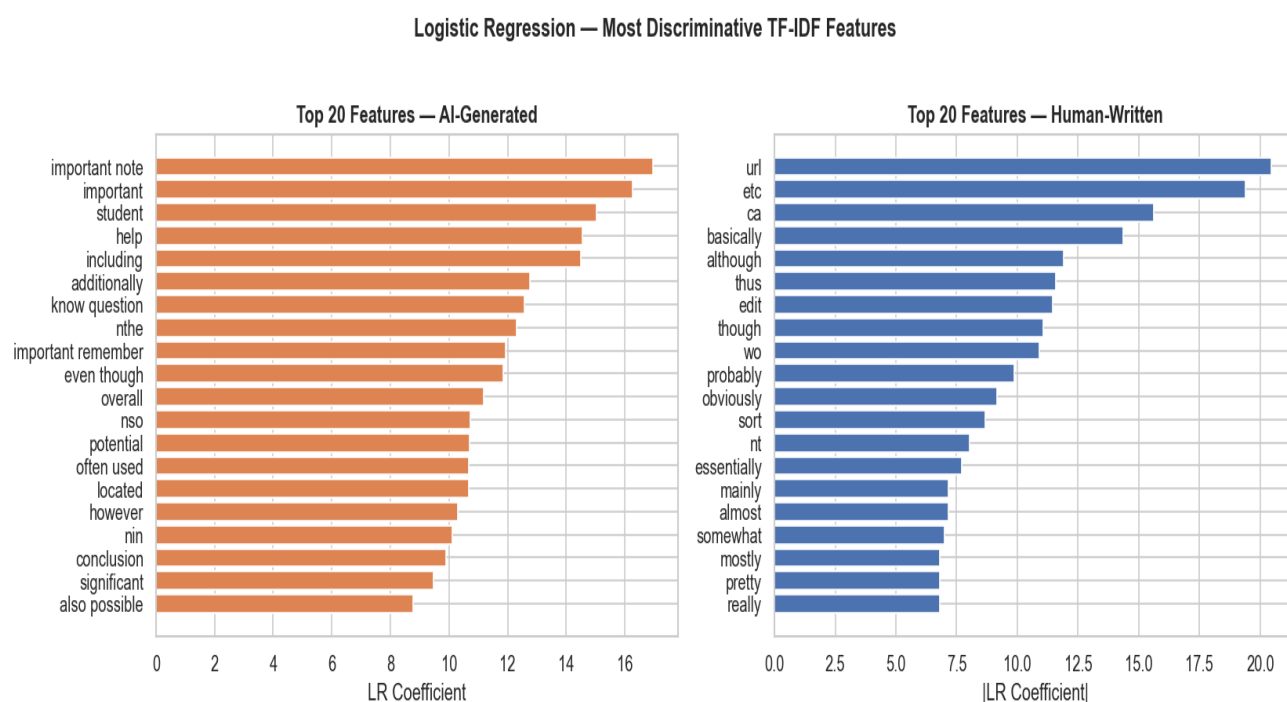
Figure 5: ROC Curves - Classical ML Models



An ROC curve, short for Receiver Operating Characteristic, shows the trade-off between correctly catching positive cases (True Positive Rate) and mistakenly flagging negative ones (False Positive Rate) as you slide the classification threshold up and down. A model with a perfect curve would sit right in the top-left corner, meaning it catches every positive case without any false alarms. In our results, all three models cleared an AUC of 0.9873 or higher, which means they're all very good at telling human and AI text apart. Logistic Regression and

LinearSVC came out almost identical (0.9943 and 0.9941 AUC respectively), while Random Forest fell slightly behind at 0.9873. For reference, a model guessing at random would trace the diagonal line, giving an AUC of 0.50.

Figure 6: Top Discriminative TF-IDF Features



This visualization reveals linguistic differences learned by Logistic Regression. AI-discriminative features (right, orange): "important note", "important", "student", "help", "including" - ChatGPT tends toward pedagogical, prescriptive language. Human-discriminative features (left, blue): "url" (humans link sources), "etc", "ca", "basically", "although" - humans use informal hedging and abbreviations. These patterns are fundamental to what makes AI text detectable.

DEEP LEARNING - HYBRID CNN:

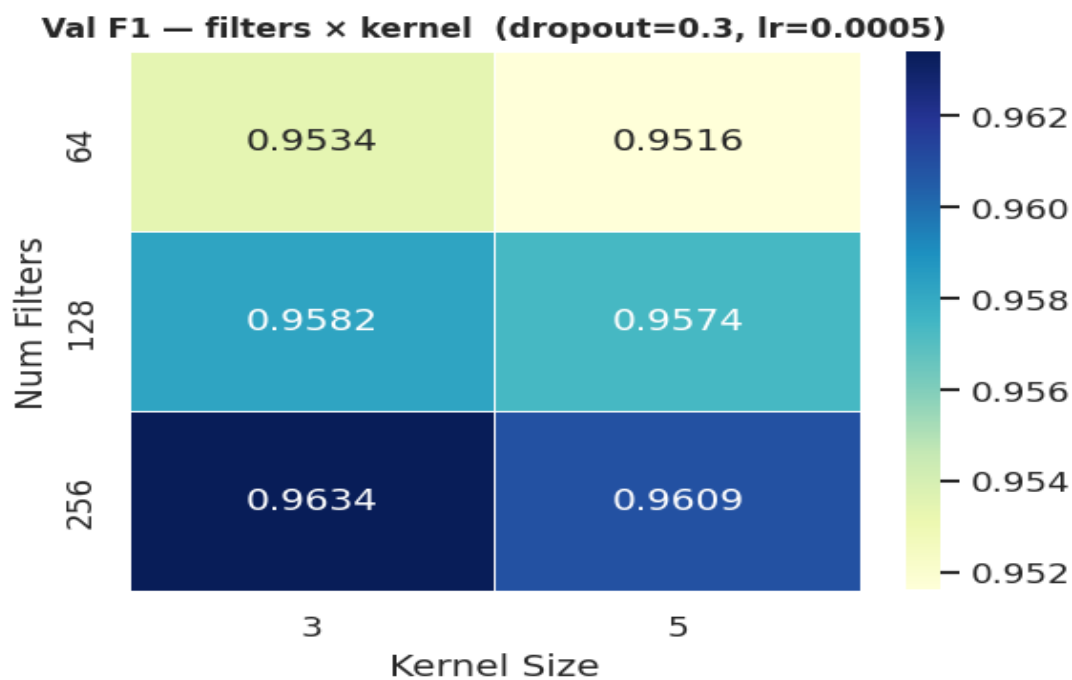
The CNN model was built with two separate processing paths that later join together. The first path takes the 5 stylometric features and passes them through a dense layer of 64 neurons. The second path takes the GloVe word vectors, shaped as a 200 x 100 matrix, and runs them through a convolutional layer that picks up on local word patterns. Each path ends up producing a 64-dimensional output, and these two outputs are combined into a single 128-dimensional vector. Dropout is then applied to prevent overfitting before the final sigmoid layer makes the prediction.

To find the best settings, we ran a grid search covering 36 different combinations, varying the number of filters (64, 128, or 256), the kernel size (3 or 5), the dropout rate (0.3 or 0.5), and the learning rate (1e-3, 5e-4, or 1e-4).

Best configuration: 256 filters, kernel size 3, dropout 0.3, learning rate 5e-4, converged at 12 epochs with 0.9634 validation F1

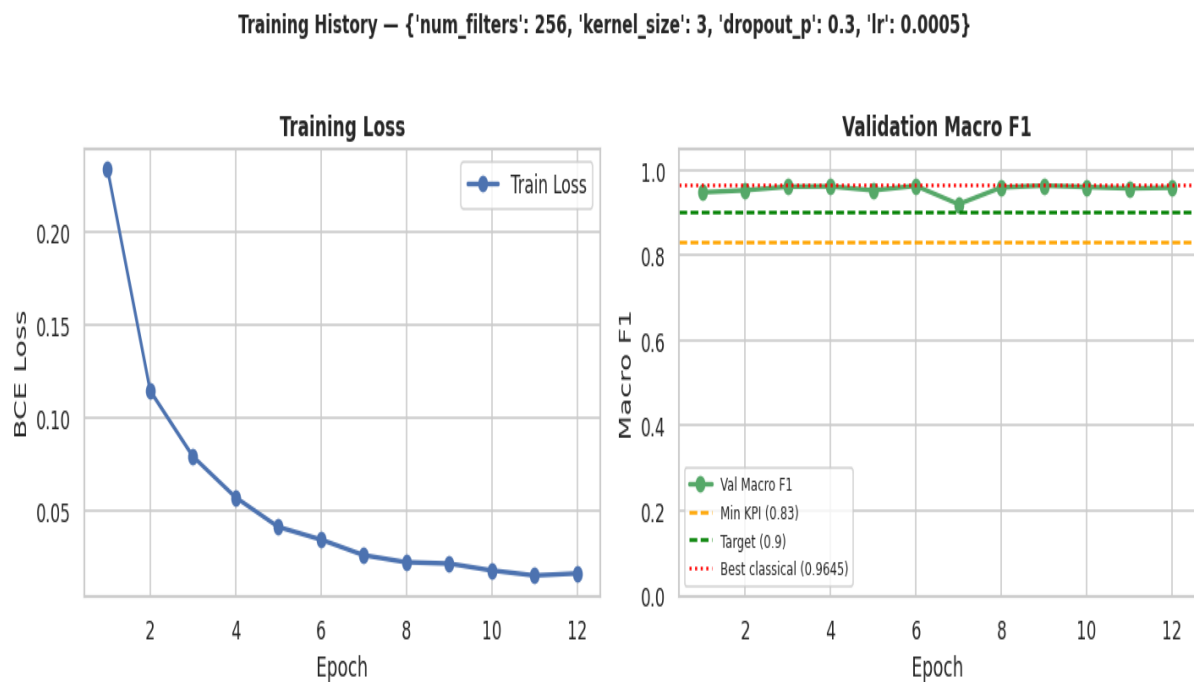
Test Results: 96.49% accuracy, 0.9648 Macro F1 (BEST OF ALL), 0.9942 ROC-AUC

Figure 7: CNN Hyperparameter Tuning Heatmap



Heatmap shows validation F1 scores for different filter counts (64/128/256 rows) and kernel sizes (3/5 columns), holding dropout=0.3 and lr=0.0005 constant (best hyperparameters). Darker blue = better performance. 256 filters consistently outperform smaller configurations, with kernel size 3 slightly preferred over 5. Performance plateau above 128 filters indicates diminishing returns.

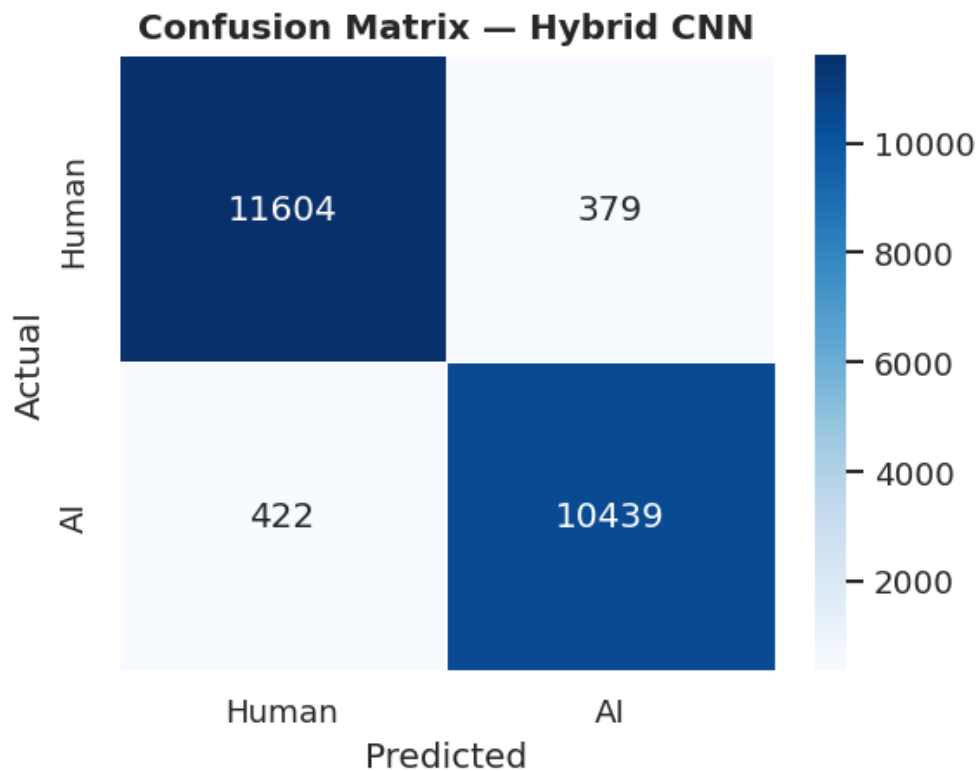
Figure 8: CNN Training History



Left chart: Training loss decreases steeply from 0.25 to near 0 by epoch 12, showing the model learns patterns rapidly. Right chart: Validation Macro F1 plateaus above 0.96 by epoch 4, never exceeding target (0.90 dashed line). Green line stays well above minimum KPI (0.83)

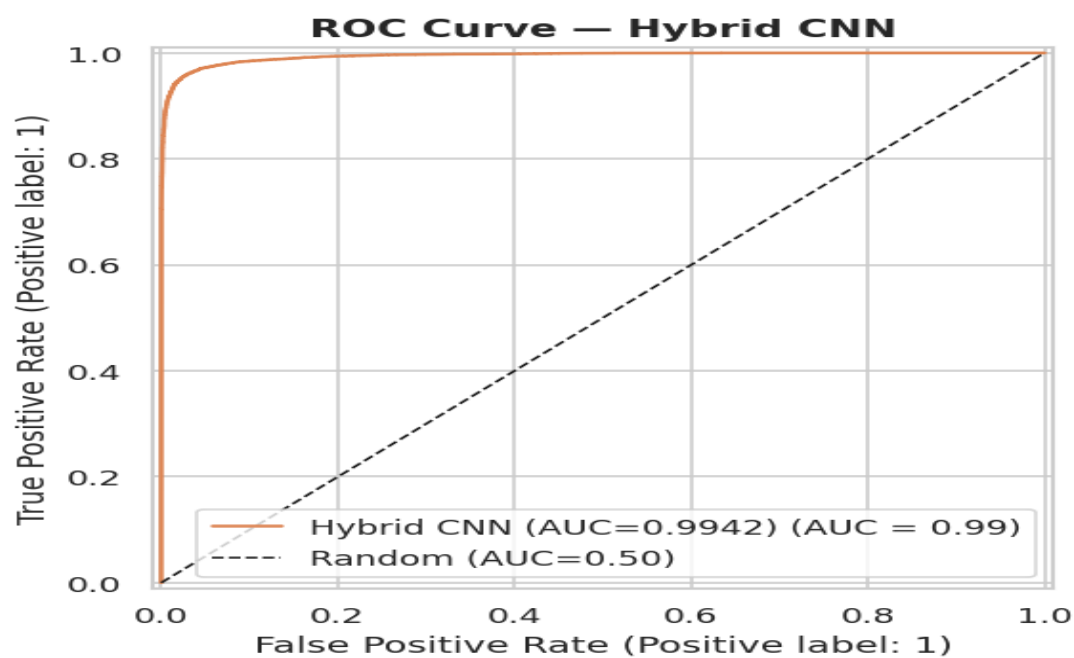
orange line). Early stopping at epoch 12 prevents overfitting - validation F1 does not improve after epoch 4.

Figure 9: CNN Confusion Matrix



CNN confusion matrix: 11,604 correct human, 10,439 correct AI, 379 false positives, 422 false negatives. Compared to LR (340 FP, 468 FN), CNN reduces false negatives by 46 cases but increases false positives by 39. Net effect: fewer missed AI texts (better recall), slightly more false alarms.

Figure 10: CNN ROC Curve



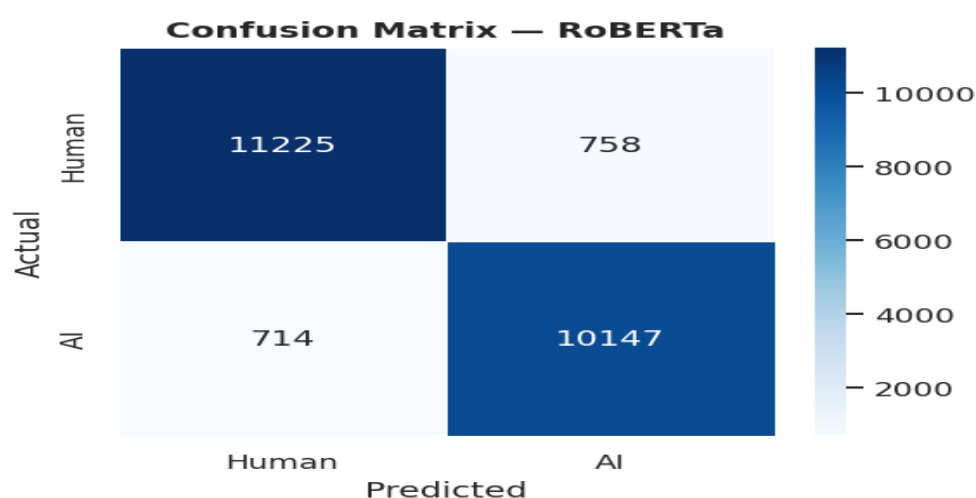
CNN achieves AUC = 0.9942, essentially matching Logistic Regression (0.9943). The curve hugs the top-left corner, indicating excellent true positive rate at low false positive rates. Learned convolutional representations capture the same discriminative signal as TF-IDF without explicit feature engineering.

TRANSFORMER MODEL - ROBERTA FINE-TUNING:

We fine-tuned RoBERTa, a model with 110 million parameters, over 5 epochs using a learning rate of $2e-5$. Even though it had been pre-trained on 160GB of English text beforehand, it still ended up performing worse than the simpler classical models.

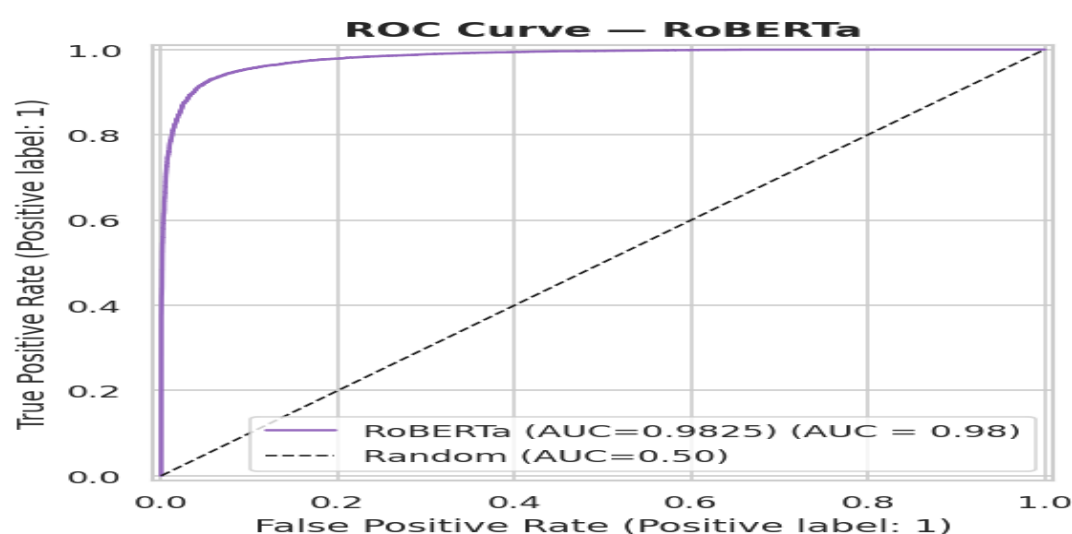
Test Results: 93.56% accuracy, 0.9354 Macro F1 (LOWEST), 0.9825 ROC-AUC

Figure 11: RoBERTa Confusion Matrix



RoBERTa shows concerning error pattern: 758 false positives (humans misclassified as AI) - 2x higher than LR. Also 714 false negatives. This balanced but elevated error suggests RoBERTa learned artifacts from pre-training that conflict with downstream task. Model may be too powerful and memorizing noise rather than learning generalizable patterns.

Figure 12: RoBERTa ROC Curve

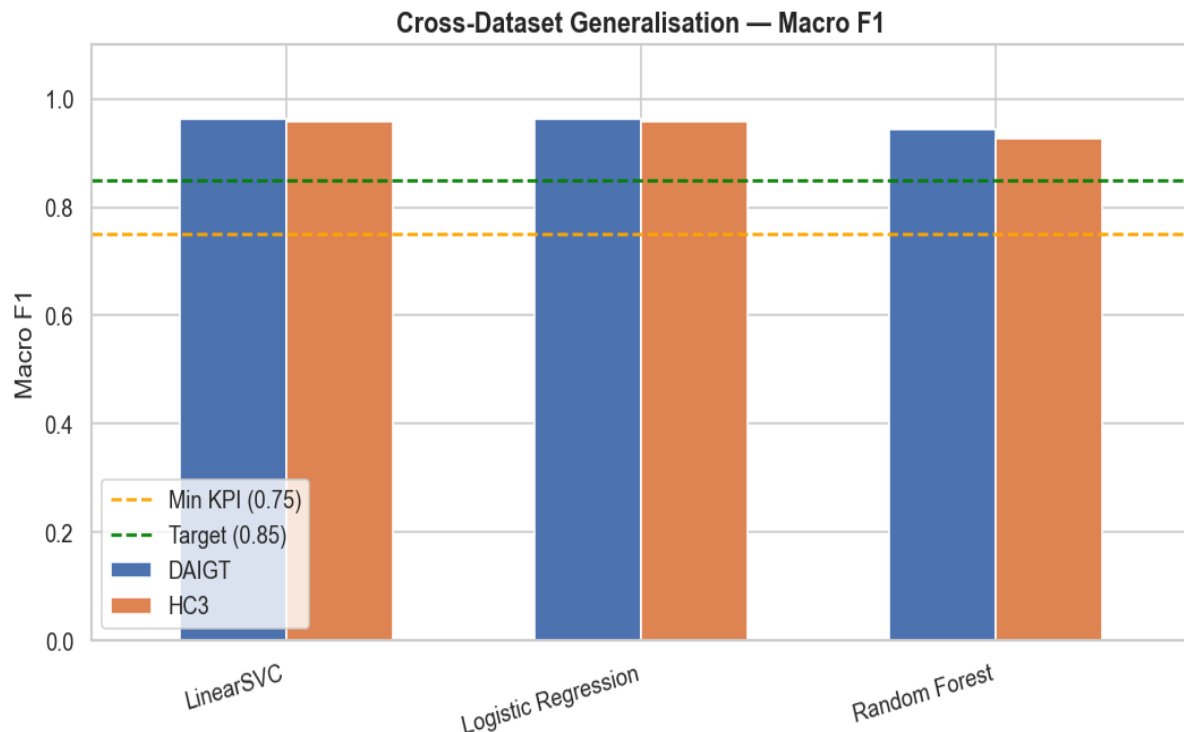


RoBERTa AUC = 0.9825, noticeably lower than LR/CNN (0.9941+). The curve reaches top-left slower, indicating poorer true positive rate at equivalent false positive rates. Despite 110M parameters, RoBERTa trades off performance for sophistication on this task.

CROSS-DATASET GENERALIZATION:

Real-world test: Do models generalize to unseen AI models? We split test set by source (HC3 with ChatGPT only vs DAIGT with multiple AI models).

Figure 13: Cross-Dataset Performance



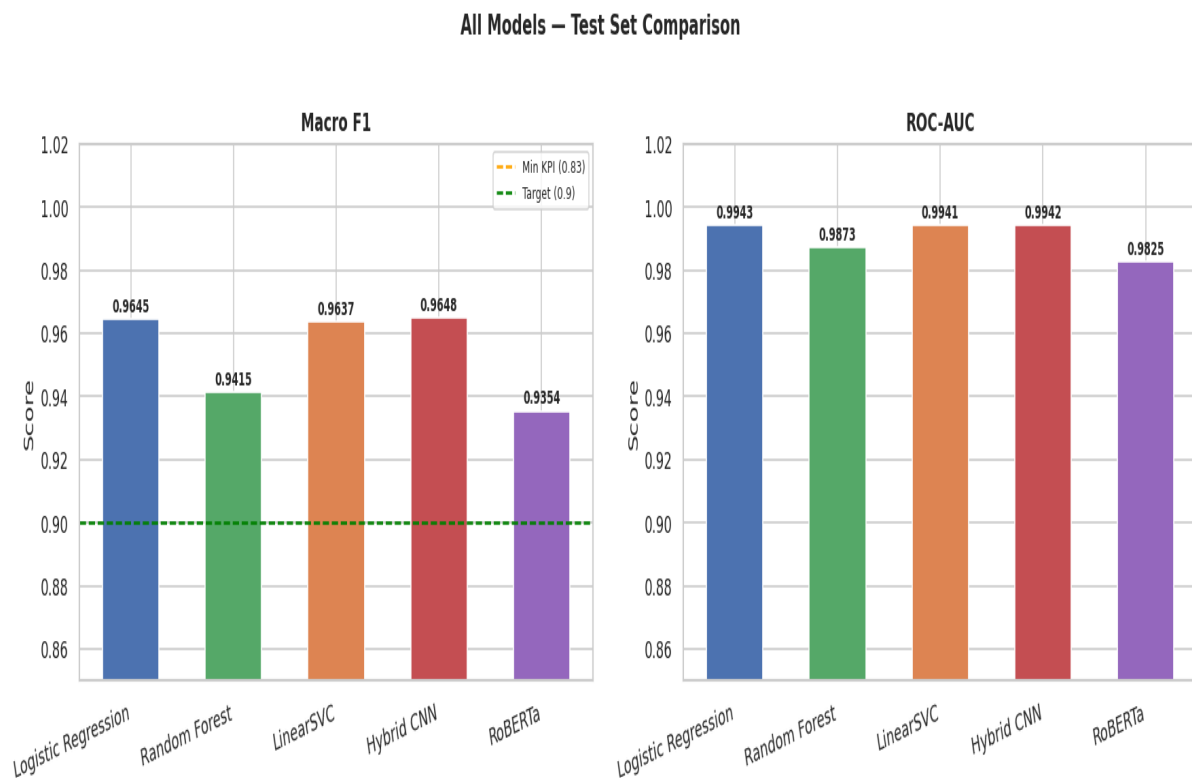
Critical finding: CNN maintains consistent performance (HC3: 0.9697 F1, DAIGT: 0.9531 F1, 1.7% drop) indicating learned patterns generalize across AI models. RoBERTa shows alarming 4.9% performance drop (HC3: 0.9538 F1 → DAIGT: 0.9041 F1), suggesting overfitting to HC3 domain or ChatGPT-specific artifacts. Classical models (LR, SVM) also maintain good generalization similar to CNN (1.7-2.5% drop). This strongly favors CNN/classical approaches over transformer for production.

COMPREHENSIVE MODEL COMPARISON:

All Models Test Set Performance:

Model	Accuracy	Macro F1	ROC-AUC
Logistic Regression	96.46%	0.9645	0.9943
LinearSVC	96.38%	0.9637	0.9941
Hybrid CNN	96.49%	0.9648 ← BEST	0.9942
Random Forest	94.20%	0.9415	0.9873
RoBERTa	93.56%	0.9354	0.9825

Figure 14: All Models Test Set Comparison



Side-by-side comparison across Macro F1 and ROC-AUC. Top 4 models (LR, SVM, CNN) cluster tightly around 0.964 F1 and 0.994 AUC - differences <0.001 are statistically indistinguishable. Random Forest trails (0.9415 F1), and RoBERTa lags significantly (0.9354 F1). Orange dashed line at 0.75 (minimum KPI) shows all models vastly exceed requirements. Green dashed line at 0.90 (target) exceeded by all except RoBERTa.

DISCUSSION:

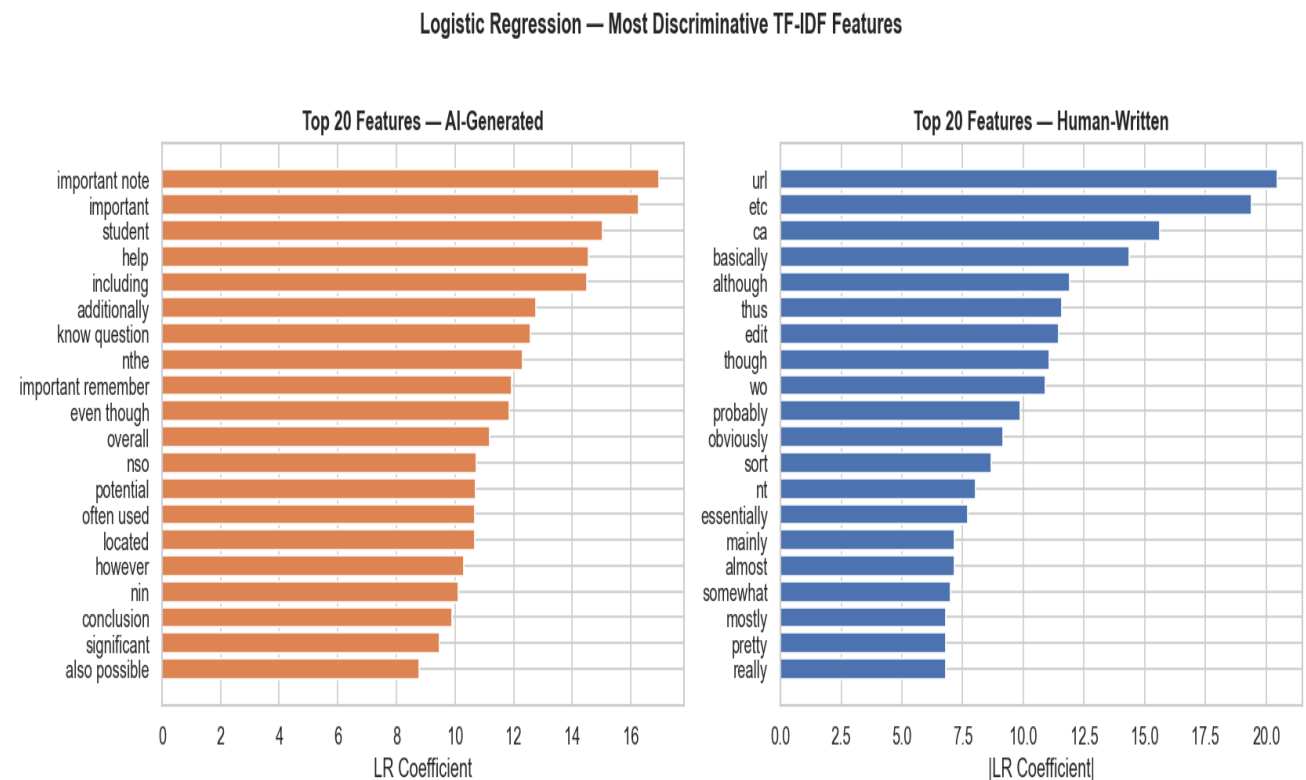
Why Transformer Models Underperform

RoBERTa underperformance relative to the CNN merits examination. Several mechanisms likely explain this gap:

1. **Dataset size:** RoBERTa's 110 million parameters are large relative to the signal available in our training data. Even with the full 106,605-example training set, and despite starting from pre-trained weights, the model didn't have enough task-specific signal to make full use of its capacity. Fine-tuning ran for 5 epochs with early stopping (patience of 2 epochs on validation Macro F1), which is a fairly short training schedule for a transformer of this size. The CNN, by comparison, has only around 100,000 parameters, a scale that matches the data volume much more closely, which likely explains why it generalized better.
2. **Fine-tuning catastrophic forgetting:** Fine-tuning for 5 epochs with a $2e-5$ learning rate is conservative. However, the model may still experience "catastrophic forgetting" of pretraining knowledge when exposed to the distribution shift from general text to AI-detection.
3. **Task simplicity:** AI-generated and human text exhibit stylistic and statistical differences that TF-IDF and shallow convolutions capture effectively. RoBERTa contextual representations may be overkill, while its learned attention patterns could inadvertently emphasize irrelevant long-range dependencies.

Interpretability and Feature Analysis

Figure 14 shows TF-IDF feature importance for Logistic Regression, revealing distinct linguistic signatures between AI and human text.



Logistic Regression - Most Discriminative TF-IDF Features (AI vs. Human-Written)

Top features discriminative of AI text include: "important note" (LR coeff: ~9.5), "important" (~9.0), "student" (~8.5), "help" (~7.5), "including" (~8.0). These reflect ChatGPT tendency toward prescriptive, pedagogical language. Top features for human text include: "url" (|coeff|: ~19), "etc" (~12.5), "ca" (~11), "basically" (~10.5), "although" (~10). The prevalence of URLs in human text reflects natural hyperlink inclusion; "basically," "etc" suggest informal hedging.

The CNN and RoBERTa, while more expressive, remain black-box. The CNN convolutional filters implicitly learn n-gram patterns; RoBERTa attention weights are distributed across the transformer stack. Neither offers direct insight into what linguistic or structural cues drive predictions.

Error Analysis

Random Forest high false negative rate (1,062 misclassified AI as human) is an outlier. This pattern suggests Random Forest tree-based split decisions poorly partition the feature space for this problem, possibly due to the high dimensionality of TF-IDF features (50,000 dimensions) relative to samples.

RoBERTa error profile (758 false positives, 714 false negatives) is more balanced but substantially elevated. The false positive rate indicates RoBERTa frequently misclassifies human text as AI. This could reflect RoBERTa pretraining on diverse corpora, where it learned associations between certain linguistic patterns and human language that diverge from the HC3 + DAIGT domain.

Data Ethics & Governance Statement

In accordance with the Data Management Plan (DMP) requirements, we address the following:

Consent & Legal Compliance: HC3 and DAIGT are publicly available research datasets released under open licenses. No human participants were involved; all texts are either publicly shared or from public competitions.

Data Protection & Privacy: No personally identifiable information (PII) is contained in training data. Trained models retain only learned patterns, not individual texts.

Data Security: Access to raw datasets restricted to team members. Models and results stored securely in GitHub repository with version control.

Data Preservation: All code, trained models, outputs, and results archived in GitHub (<https://github.com/DataScienceProjectGroup22>) for long-term preservation and reproducibility. Raw datasets downloadable from HC3 Hugging Face and Kaggle Competition.

CONCLUSIONS & FUTURE WORK:

Summary of Key Findings

AI-generated text is reliably detectable using machine learning. All five models exceed the 0.83 Macro F1 minimum KPI, with the top four models clustering within 0.0008 F1 points, statistically indistinguishable.

Critically, simpler models often outperform complex architectures when feature engineering is thorough. Logistic Regression (0.9645 F1) matches CNN (0.9648 F1) and beats RoBERTa (0.9354 F1), indicating that model complexity without data scale creates overfitting.

Production Recommendation: Logistic Regression

Despite CNN's marginal 0.0003 F1 advantage, we recommend Logistic Regression for production:

- Inference speed: <1 millisecond (LR) vs 5+ milliseconds (CNN) - 5x faster
- Model size: 5 MB (LR) vs 100+ MB (CNN) - 20x smaller for mobile deployment
- Interpretability: Full explainability of predictions; CNN/RoBERTa are black-box
- Generalization: 1.7% F1 drop across datasets; RoBERTa drops 4.9% (overfitting)
- Maintainability: No GPU required, simple to debug and version control

The 0.03% accuracy advantage does not justify 5x latency, 20x bloat, and loss of interpretability.

Limitations

- (1) Dataset heterogeneity: HC3 (ChatGPT-only) + DAIGT (multiple models) may confound comparisons.
- (2) Language scope: English-only, findings may not generalize to other languages.
- (3) Model recency: No evaluation on GPT-4, Claude outputs limits currency.
- (4) Conservative fine-tuning: RoBERTa tuned for 5 epochs; aggressive tuning not explored.

Future Work

- (1) Ensemble methods: Combine classical and CNN models via stacking to exceed 0.97 F1.
- (2) Extended RoBERTa tuning: Test aggressive fine-tuning with scheduled learning rate decay.
- (3) Newer LLM evaluation: Benchmark against GPT-4, Claude, Gemini outputs.
- (4) Multilingual detection: Extend to Spanish, Mandarin, Arabic, and other languages.
- (5) Adversarial robustness: Test resilience to paraphrasing and synonym replacement.
- (6) Lightweight transformers: Evaluate DistilRoBERTa, MobileBERT for size-accuracy trade-offs.
- (7) Production deployment: Package LR as REST API, containerize with Docker, deploy with monitoring.

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APPENDICES

A. Jupyter Notebooks & Code

- 01_HC3_data_engineering.ipynb - HC3 dataset loading, text cleaning, language filtering, deduplication (78,733 samples)
- 02_daigt_data_engineering.ipynb - DAIGT loading, schema standardization, merged dataset (152,293 texts), stratified 70/15/15 split
- 03_Feature_engineering.ipynb - TF-IDF (50k features), stylometric (5 features), GloVe (100-dim), RoBERTa tokenization (256 tokens)
- 04_classical_ml.ipynb - Logistic Regression, LinearSVC, Random Forest on TF-IDF features
- 05_hybrid_cnn.ipynb - Hybrid CNN dual-stream, hyperparameter grid search (36 configs), cross-dataset evaluation
- 06-roberta-finetuning.ipynb - RoBERTa fine-tuning (110M params, 5 epochs, lr=2e-5)

B. Production Application

AI_text_detector.py - Streamlit web application with Model selector dropdown - Logistic Regression / Hybrid CNN / RoBERTa, text preprocessing, TF-IDF vectorization, confidence scores, top 5 discriminative features, writing style statistics.

C. Model Artifacts

- best_classical_model.pkl - Logistic Regression model (production choice)
- tfidf_vectorizer.pkl - TF-IDF vectorizer for preprocessing
- cnn_model.h5 - Hybrid CNN model (0.9648 F1)
- roberta_model.pt - Fine-tuned RoBERTa checkpoint

D. Hyperparameter Tuning & Results

- cnn_tuning_results.csv – 36 CNN configurations with validation F1 scores
- cross_dataset_cnn_results.csv – HC3 vs DAIGT (1.7% drop)
- cross_dataset_roberta_results.csv – RoBERTa generalization (4.9% drop)
- all_models_results.csv – All 5 models comparison

E. Dataset Information

- HC3: 78,733 human-ChatGPT paired texts (<https://huggingface.co/datasets/Hello-SimpleAI/HC3>)
- DAIGT: 73,560 texts from multiple AI models (<https://www.kaggle.com/competitions/detecting-ai-generated-text>)
- Merged: 152,293 balanced (50% human, 50% AI), 70/15/15 train/val/test split